1 Research project's background

This project studies how external uncertainty shocks originating in the United States and China are transmitted to Taiwan, which external sources matter most for Taiwan's domestic uncertainty, and whether the dominant transmission channel changes over time. These questions are especially important for Taiwan because it is deeply integrated with both the U.S. and Chinese economies. As a result, Taiwan lies directly in the path of major policy and geopolitical developments between the two countries. When the U.S. Federal Reserve tightens monetary policy, when trade disputes between the U.S. and China escalate, or when cross-strait political tensions rise, Taiwan often faces immediate consequences for its trade, investment, and financial flows.

However, major policy shifts and geopolitical realignments rarely move from initial proposal to a well-defined policy trajectory overnight. Central banks move in sequences of decisions, trade conflicts unfold through rounds of negotiation and retaliation, and cross-strait relations evolve over extended periods rather than at a single point in time. As a result, there is often a long interval between the announcement or anticipation of a policy change and the point at which a stable policy regime actually emerges. Along the way, planned policy changes may be revised, delayed, or even cancelled altogether. For example, rounds of U.S.–China tariff threats have repeatedly ended with smaller or postponed tariff increases than initially announced, and cross-strait initiatives have at times been shelved after domestic political opposition. During this interval, firms, households, and investors face heightened uncertainty about the future path of policy, demand, and financial conditions. This heightened uncertainty, in turn, affects their investment, consumption, and portfolio decisions (Bloom, 2009, 2014). One prominent example is trade policy uncertainty, which strongly influences firms' trade and investment decisions (Handley and Limao, 2022). To capture these evolving dynamics without sacrificing the macroeconomic dimension, this project utilizes a monthly dataset. Monthly data are granular enough to capture short-lived spikes in uncertainty during negotiation periods that quarterly data often smooth out, while still allowing the analysis to include key real variables such as industrial production and trade balances that are generally unavailable at daily frequency.

Empirically, however, identifying external uncertainty shocks and quantifying their effects on

Taiwan presents several methodological challenges. First, external and domestic sources of uncertainty are tightly intertwined, creating a severe challenge for identification. A single geopolitical event, such as a shift in U.S.–China relations, often simultaneously triggers global financial volatility and shifts in Taiwan's domestic political landscape. In a small-scale model that lacks sufficient domestic controls, the economic impact of these internal political shifts would be erroneously attributed solely to the external shock. This form of omitted variable bias can lead to a misdiagnosis of the true drivers of uncertainty, potentially overstating the direct influence of external factors while neglecting local transmission mechanisms (Carriero, Clark and Marcellino, 2018). Addressing this issue requires expanding the information set to include a rich array of both domestic and external variables, necessitating the use of a large-scale econometric model.

Second, while a large-scale model is necessary to address this bias, it introduces identification trade-offs. In large systems, standard identification methods based on Cholesky decomposition become problematic because the results are sensitive to variable ordering, creating a conceptual flaw known as order dependence. Addressing this issue requires an order-invariant framework. However, order invariance is mathematically incompatible with the strict block-exogeneity restrictions commonly used in small open economy models to prevent domestic variables from affecting global ones. Consequently, instead of imposing these rigid constraints, we employ the data-driven identification strategy proposed by Davidson, Hou and Koop (2025), treating external drivers as "unclassified variables". This avoids arbitrary restrictions on the contemporaneous relationships between variables, allowing for a data-determined structure that can accommodate potential feedback loops, rather than ruling them out by assumption. This approach suits Taiwan's pivotal role in global technology supply chains, where strict zero restrictions might assume away significant feedback effects originating from the supply side.

Third, many existing approaches face two related limitations. One is their reliance on proxy measures of uncertainty. The other is their use of a two-step procedure that first estimates uncertainty and then evaluates its macroeconomic effects in a separate model. Regarding the former limitation, widely used text-based indices built from international news may not reflect domestic conditions. For example, during Nancy Pelosi's 2022 visit, international coverage emphasized imminent war risk, while sentiment in Taiwan remained relatively calm. Such reliance on external proxies can create a "perception gap" and distort estimates of uncertainty's impact on the local economy. Regarding the latter limitation, the two-step design treats the estimated uncertainty series as observed data, ignores estimation uncertainty, and can induce measurement-error bias and model inconsistency (Carriero, Clark and Marcellino, 2018). To address these issues, we employ the proposed stochastic volatility in mean vector autoregression (SVMVAR) framework to jointly estimate uncertainty and its economic effects in a single step. By modeling uncertainty as a latent factor driven by the data itself, we avoid reliance on potentially misaligned external proxies, ensuring that our measure reflects actual domestic economic and financial conditions.

Finally, existing uncertainty measures, such as global financial volatility indices or country-specific policy uncertainty indices, do not distinguish between the *channels* through which uncertainty transmits to the economy. Widely adopted indices such as the VIX or the Baker, Bloom and Davis (2016) Economic Policy Uncertainty index compress the multidimensional nature of uncertainty into a single scalar, obscuring whether a given shock propagates through real activity or financial markets. For a small open economy like Taiwan, this distinction carries different policy implications. When external shocks operate primarily through **macroeconomic channels**, affecting trade flows, export demand, and production linkages, the appropriate policy response involves instruments oriented toward the real economy, such as export facilitation and structural adjustment support. When shocks instead propagate through **financial channels**, disrupting capital flows, asset prices, and credit conditions, the Central Bank of Taiwan faces a different set of imperatives, including foreign exchange intervention and liquidity management. Existing empirical frameworks, however, are ill-equipped to make this distinction in a time-varying setting, leaving policymakers without a systematic basis for identifying which channel is dominant at any given point in time. This constitutes the fourth and most substantive methodological gap that the present project is designed to fill.

To address these challenges, this project applies the order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) framework developed by Davidson, Hou and Koop (2025) to the Taiwan context. This project treats external drivers—such as U.S. monetary policy indicators and cross-strait tension indices—as unclassified variables and uses their time-varying co-movement with Taiwan's macroeconomic and financial common factors to identify the operative transmission channel at each point in time without requiring the researcher to impose it ex ante.

While the OI-SVMVAR identifies the dominant transmission channels and estimates both macroeconomic and financial uncertainty shocks, it is silent on the underlying micro-founded frictions driving these responses. In other words, researchers need to determine exactly where the second-moment shocks should be introduced into the theoretical model. Existing dynamic stochastic general equilibrium (DSGE) literature often relies on ad hoc assumptions, arbitrarily placing volatility shocks on productivity, borrowing constraints, or investment processes without empirical justification.

To bridge this gap, this project develops a novel nonlinear uncertainty accounting framework. By extending the Business Cycle Accounting (BCA) methodology to a monetary small open economy via higher-order perturbation, this project removes the need for ad hoc shock placement. Instead, we use a flexible factor-loading structure to endogenously map the empirically extracted uncertainty factors from Year 1 into the volatility processes of the structural wedges. This provides a rigorous, data-driven mechanism to pinpoint exactly where external uncertainty enters and distorts the domestic economy—whether through cross-border risk premium spikes or domestic investment delays—offering a micro-founded validation of our OI-SVMVAR results.

This project makes four contributions. First, the large-scale specification, which includes more than forty domestic, U.S., Chinese, and global variables, helps mitigate the omitted-variable bias that is especially severe when external and domestic sources of uncertainty are tightly intertwined (Carriero, Clark and Marcellino, 2018). Second, the order-invariant identification strategy avoids the distortions associated with variable ordering in large VAR systems. Third, and most importantly for policy, the model's time-varying classification results help identify whether policymakers in Taiwan should respond primarily with real-economy instruments or with financial stability tools, a distinction that conventional single-index measures of uncertainty cannot provide. Fourth, the novel nonlinear uncertainty accounting framework bridges the empirical and theoretical literature, providing a generalized, micro-founded diagnostic tool to evaluate the real effects of second-moment shocks without relying on arbitrary shock placement.

The remainder of this proposal proceeds as follows. The first year of the project focuses on data assembly and empirical implementation: we construct a monthly dataset of more than forty variables spanning Taiwan's domestic macroeconomic and financial conditions together with U.S., Chinese, and global indicators, adapt the Davidson, Hou and Koop (2025) MCMC estimation algorithm to this dataset, and conduct the three-step analysis comprising time-varying classification probabilities, forecast error variance decomposition (FEVD), and impulse response functions (IRF). The second year extends the analysis by developing a monetary small open economy New Keynesian model to execute a novel nonlinear uncertainty accounting analysis. Solved via third-order perturbation and estimated using Bayesian methods, this structural framework maps the data-driven uncertainty factors extracted in the first year into the stochastic volatility processes of structural wedges, providing a rigorous, micro-founded validation of the transmission mechanisms identified empirically.

The First Year

1.1 Methods, procedures, and implementation schedule

(1) Research principles, methods, and the innovation of research methods

(1.1) The OI-SVMVAR Framework

To address the identification challenges outlined in Section 1, this project employs the **order-invariant stochastic volatility in mean vector autoregression** (OI-SVMVAR) framework developed by Davidson, Hou and Koop (2025). We select this framework because it uniquely resolves the three methodological obstacles that confront any attempt to trace external uncertainty shocks through a small open economy: it accommodates a large information set, eliminating the omitted variable bias that confounds small-scale models; its order-invariant identification ensures that results do not depend on arbitrary variable ordering; and its time-varying classification mechanism allows the data—rather than the researcher—to determine whether a given external shock transmits through macroeconomic or financial channels. The model consists of five coupled components, estimated jointly in a single Bayesian step, which we now present in turn.

To capture the direct impact of uncertainty on real economic activity and financial conditions, the observation equation embeds the latent uncertainty factors in the conditional mean of a VAR:

$$\mathbf{y}_t = \mathbf{c} + \sum_{l=1}^{p} \mathbf{A}_l \, \mathbf{y}_{t-l} + \boldsymbol{\Gamma} \, \mathbf{h}_t + \boldsymbol{\varepsilon}_t, \tag{1}$$

where $\mathbf{y}_t$ is the $N \times 1$ vector of observables (in our application, $N = 45$ variables spanning Taiwan's domestic economy, U.S. and Chinese indicators, and global risk measures); $\mathbf{c}$ is an $N \times 1$ intercept vector; $\mathbf{A}_l$ are $N \times N$ coefficient matrices at lag $l = 1, \ldots, p$; $\boldsymbol{\varepsilon}_t$ is the $N \times 1$ reduced-form disturbance vector; and $\mathbf{h}_t = (h_{m,t},\, h_{f,t})'$ is a 2×1 vector containing the two latent common log-volatility factors that represent Taiwan's **macroeconomic uncertainty** ($h_{m,t}$) and **financial uncertainty** ($h_{f,t}$), respectively. The $N \times 2$ matrix $\boldsymbol{\Gamma}$ captures the **stochastic volatility in mean** effect: because $\boldsymbol{\Gamma} \, \mathbf{h}_t$ enters the conditional mean of $\mathbf{y}_t$, elevated uncertainty directly shifts the expected path of all observables—output, employment, asset prices, and credit conditions alike. This is what distinguishes the SVMVAR from a standard stochastic volatility model, in which volatility appears only in the error covariance and thus cannot feed back into the levels of economic variables. The inclusion of $\boldsymbol{\Gamma} \, \mathbf{h}_t$ allows us to test the prediction—central to real-options and precautionary-savings theories (Bloom, 2009, 2014)—that heightened uncertainty depresses investment, consumption, and hiring not merely by increasing dispersion but by shifting the conditional mean of economic outcomes.

To ensure that identification does not depend on the arbitrary ordering of variables within $\mathbf{y}_t$, the contemporaneous relationship between reduced-form and structural disturbances is specified as

$$\mathbf{B}_0 \, \boldsymbol{\varepsilon}_t = \boldsymbol{\Sigma}_t^{1/2} \, \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \, \mathbf{I}_N), \tag{2}$$

where $\mathbf{B}_0$ is an $N \times N$ contemporaneous impact matrix and $\boldsymbol{\Sigma}_t = \mathrm{diag}(\sigma^2_{1,t}, \ldots, \sigma^2_{N,t})$ is the diagonal matrix of time-varying, variable-specific conditional variances. The structural disturbances $\mathbf{u}_t$ are homoskedastic by construction; all time variation in volatility is absorbed by $\boldsymbol{\Sigma}_t$. Crucially, $\mathbf{B}_0$ is left as a full, unrestricted nonsingular matrix—subject only to the normalisation that its diagonal elements equal one—rather than being constrained to a lower triangular form. This unrestricted specification is what delivers **order invariance**: because no zero restrictions are imposed on the off-diagonal elements of $\mathbf{B}_0$, the estimation results are identical regardless of how variables are ordered in $\mathbf{y}_t$. The implications of this design choice—and why it is indispensable for a 45-variable system that mixes domestic and external series—are discussed in detail in subsection (1.3).

The structural heart of the model lies in the specification linking each variable's log-volatility to the common uncertainty factors. Following Davidson, Hou and Koop (2025), the N variables in $\mathbf{y}_t$ are partitioned into three blocks—n_m macroeconomic variables, n_f financial variables, and n_u unclassified variables, with $n_m + n_f + n_u = N$—and the log conditional variance of variable i is decomposed into a **variable-specific idiosyncratic component** $\eta_{i,t}$ and a loading on the relevant common factor:

$$\log \sigma_{i,t}^2 = \begin{cases} \eta_{i,t}^m + h_{m,t} & \text{if variable } i \text{ is macroeconomic,} \quad i = 1, \dots, n_m, \\ \eta_{i,t}^f + h_{f,t} & \text{if variable } i \text{ is financial,} \quad i = 1, \dots, n_f, \\ \eta_{i,t}^u + h_{s_{i,t},t} & \text{if variable } i \text{ is unclassified,} \quad i = 1, \dots, n_u. \end{cases} \tag{3}$$

The idiosyncratic terms $\eta_{i,t}^k$ ($k \in \{m, f, u\}$) capture variable-specific volatility movements that are orthogonal to the common factors, ensuring that the log-volatilities of variables within the same block are not forced into perfect collinearity. Each $\eta_{i,t}^k$ follows a stationary AR(1) process, so that idiosyncratic volatility shocks are mean-reverting while the common factors $h_{m,t}$ and $h_{f,t}$ can exhibit the persistent swings characteristic of aggregate uncertainty episodes. Variables pre-classified as macroeconomic load exclusively on $h_{m,t}$, so that their common volatility component rises and falls only with aggregate real-economy uncertainty. Variables pre-classified as financial load exclusively on $h_{f,t}$. For unclassified variables, however, the common factor is selected by a discrete latent state indicator $s_{i,t} \in \{m, f\}$: when $s_{i,t} = m$, the variable's common volatility component is $h_{m,t}$; when $s_{i,t} = f$, it is $h_{f,t}$. This discrete switching mechanism—rather than a continuous blend of both factors—means that at each point in time the model assigns each unclassified variable entirely to one block, and the *posterior probability* of that assignment becomes the object of inferential interest. As we explain in subsection (1.2), this is the mechanism through which we identify whether a given external uncertainty source transmits to Taiwan through the macroeconomic channel or the financial channel.

The two common log-volatility factors evolve according to independent driftless random walks:

$$h_{k,t} = h_{k,t-1} + \zeta_{k,t}, \qquad \zeta_{k,t} \sim \mathcal{N}(0,\, \sigma^2_{\zeta,k}), \qquad k \in \{m, f\}, \tag{4}$$

where the innovation variance $\sigma^2_{\zeta,k}$ controls the degree of time variation in factor k. The random-walk specification permits persistent, unbounded movements in the log-volatility factors, consistent with the empirical evidence that uncertainty exhibits prolonged episodes of elevation—such as the 2008–2009 global financial crisis or the 2018–2019 U.S.–China trade war—followed by gradual mean reversion (Bloom, 2014). Moreover, the random walk nests the constant-volatility case as a limiting special case when $\sigma^2_{\zeta,k} \to 0$, so that the degree of time variation is estimated from the data rather than imposed by assumption. Because $h_{m,t}$ and $h_{f,t}$ are defined in logs, the implied level of uncertainty $\exp(\frac{1}{2}h_{k,t})$ is guaranteed to be positive at all times.

The final component governs the time-varying classification of unclassified variables, which is the mechanism that transforms the statistical model into an economic identification device. For each unclassified variable i, the latent state indicator $s_{i,t} \in \{\text{macro, financial}\}$ in Equation (3) follows a first-order two-state **Markov chain** with transition probabilities $q^{(i)}_{mm} = P(s_{i,t} = m \mid s_{i,t-1} = m)$ and $q^{(i)}_{ff} = P(s_{i,t} = f \mid s_{i,t-1} = f)$. The posterior classification probability

$$\pi_{i,t} = P(s_{i,t} = \text{macro} \mid \mathcal{F}_t) \tag{5}$$

is the probability—conditional on the full information set $\mathcal{F}_t$—that unclassified variable i belongs to the macroeconomic block at time t, updated recursively using the Hamilton (1989) filter. This specification implies that a variable may be classified as macroeconomic during one episode and as financial during another, with the timing and persistence of regime switches determined entirely by the data. The transition probabilities $q^{(i)}_{mm}$ and $q^{(i)}_{ff}$ are estimated jointly with all other model parameters within the Bayesian MCMC framework, so that the persistence of each variable's classification is itself a model output rather than a researcher-imposed restriction. In our application, the time path of $\pi_{i,t}$ for each external variable constitutes the primary object of interest: it reveals, at each point in the sample, whether the uncertainty associated with a given external driver transmits to Taiwan predominantly through the macroeconomic channel ($\pi_{i,t} \approx 1$) or through the financial channel ($\pi_{i,t} \approx 0$).

The full model—comprising Equations (1)–(5)—is estimated jointly in a single-step Bayesian Markov chain Monte Carlo (MCMC) procedure developed by Davidson, Hou and Koop (2025). The algorithm builds on the computationally efficient precision-based sampler of Cross et al. (2023), which exploits band and sparse matrix structures to draw the common log-volatilities in a single block, making estimation of very large SVMVARs feasible without resorting to dimensionality reduction techniques that would obscure the specific structural shocks we aim to identify. Davidson, Hou and Koop (2025) extend this approach with a novel parameter transformation for sampling the unrestricted $\mathbf{B}_0$ and with the Markov-switching classification mechanism described above. We do not derive the prior distributions or the posterior sampling steps here; the complete Bayesian implementation is detailed in the online appendix of Davidson, Hou and Koop (2025). All subsequent analysis in this project—time-varying classification probabilities, forecast error variance decompositions, and impulse response functions—is conducted on draws from the joint posterior distribution, ensuring that parameter uncertainty and classification uncertainty are fully propagated through to the final inferential objects.

(1.2) Variable Classification and the Key Innovation

The weight that the OI-SVMVAR places on its classification trichotomy—macroeconomic, financial, and unclassified—demands that the assignment of variables to the three blocks be dictated by the economic question at hand rather than by statistical convenience. Equation (3) imposes a clean mapping between blocks and the two common log-volatility factors: pre-classified macroeconomic variables load exclusively on $h_{m,t}$; pre-classified financial variables load exclusively on $h_{f,t}$; and unclassified variables load on a time-varying mixture of the two, governed by the posterior probability $\pi_{i,t}$ defined in Equation (5). Our dataset comprises $N = 45$ monthly series partitioned into $n_m = 19$ macroeconomic variables, $n_f = 10$ financial variables, and $n_u = 16$ unclassified variables. The full list of series, together with their assigned block, is reported in Table 1.

Table 1: Variable classification: 45 monthly series, 2000M1–2025M12.

Macroeconomic (19)	**Financial (10)**	**Unclassified (16)**
Taiwan real sector	*Taiwan financial markets*	*External drivers*
Industrial Production Index	Overnight Interbank Rate	*U.S. block*
Manufacturing Production Index	10-Year Govt. Bond Yield	Federal Funds Rate (effective)
Export Orders Index	Term Spread (10yr−overnight)	Industrial Production (YoY)
Retail Sales (real)	TAIEX Monthly Return	BAA–AAA Credit Spread
Exports (customs, real USD)	TAIEX Monthly Realized Volatility	Economic Policy Uncertainty
Imports (customs, real USD)	Foreign Inst. Net Buying (NTD bn)	Regional Trade Policy Uncertainty
Manufacturing PMI	Credit Spread (5yr A−5yr govt)	*China block*
Non-Manufacturing PMI	Bank Total Loans (YoY)	Industrial Production (YoY)
Business Cycle Indicator	NTD/USD Exchange Rate (nominal)	Producer Price Index (YoY)
CPI (YoY)	NTD/USD Realized Volatility	Total Social Financing (YoY)
Core CPI (YoY)		Economic Policy Uncertainty
WPI (YoY)		*Global indicators*
Import Price Index (YoY)		VIX (monthly average)
Export Price Index (YoY)		Global Geopolitical Risk
Unemployment Rate (SA)		Global Economic Policy Uncertainty
Manufacturing Employment (SA)		Taiwan Strait Geopolitical Risk
Services Employment (SA)		*U.S.–China relations*
Real Manufacturing Wages (YoY)		U.S.–China Trade Policy Uncertainty
M2 Money Supply (YoY)		U.S.–China Bilateral Trade (YoY)
		U.S.–China Bilateral GPR

Crucially, every external driver—whether a U.S. variable, a Chinese variable, a global risk indicator, or a bilateral U.S.–China measure—is assigned to the unclassified block, while both Taiwan-domestic blocks consist entirely of clearly classified variables. This allocation is not a stylistic choice but the empirical core of the research design. The transmission channel of external uncertainty shocks—whether they propagate to Taiwan through real economic activity or through financial markets—is precisely the object that this project sets out to identify. Pre-assigning the

U.S. Federal Funds Rate to the financial block, or China's Industrial Production Index to the macroeconomic block, would hardwire the answer into the prior structure of the model: the very question of whether U.S. monetary tightening reaches Taiwan through export demand or through exchange-rate and credit-market pressure would be foreclosed by assumption. By placing these variables in the unclassified block instead, the model allows the Markov-switching probability $\pi_{i,t}$ in Equation (5) to determine the operative transmission channel endogenously, at each point in time, based on how each external series co-moves with Taiwan's two common log-volatility factors in the data. The 19 macroeconomic and 10 financial variables retain their conventional classifications because their economic nature is unambiguous—industrial production is a real-sector series, the overnight interbank rate is a financial-market series—and this classification provides the anchoring structure that pins down the interpretation of $h_{m,t}$ and $h_{f,t}$ as Taiwan's domestic macroeconomic and financial uncertainty, respectively. Without such anchors, the two common factors would be statistically identified but economically interchangeable; with them in place, the loadings of the 16 external drivers acquire a direct substantive reading.

To see why $\pi_{i,t}$ functions as a transmission-channel identifier rather than as a conventional regime-switching indicator, consider the U.S. Federal Funds Rate. In a given month, the posterior probability $\pi_{i,t}$ that this variable belongs to the macroeconomic regime takes a value in $[0, 1]$. When $\pi_{i,t} \to 1$, the model's posterior assigns the Federal Funds Rate's common volatility component to $h_{m,t}$: innovations in U.S. monetary policy uncertainty co-move most strongly with Taiwan's real-sector uncertainty, indicating that the transmission occurs primarily through real channels—suppressed export demand, contracted industrial production, delayed investment, and softer manufacturing employment. When $\pi_{i,t} \to 0$, the same shock instead loads onto $h_{f,t}$: its effects operate through financial channels such as capital outflow pressure, NTD/USD depreciation, credit-spread widening, and heightened equity-market volatility. More precisely, because $\pi_{i,t}$ is a time-varying posterior probability rather than a fixed parameter, the same external variable may switch channels across episodes, with the switching dynamics governed by the variable-specific transition probabilities $q^{(i)}_{mm}$ and $q^{(i)}_{ff}$ that are estimated from the data. Time-series plots of $\pi_{i,t}$ for each of the 16 external variables therefore constitute a direct, data-driven narrative of how transmission mechanisms have evolved over the sample period—particularly across the key episodes of the 2008–2009 Global Financial Crisis, the 2015 China slowdown, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 shock, and the 2022 Federal Reserve tightening cycle. It becomes possible, for the first time in the Taiwan literature, to ask and answer whether a given external shock struck the island through the real side of the economy or through its financial markets, and whether the operative channel in 2022 was the same as the one in 2008.

This reorientation of the unclassified-variables device stands in sharp contrast to its original deployment by Davidson, Hou and Koop (2025). In their application to the U.S. economy, the unclassified block was populated by domestic series whose macroeconomic or financial nature is genuinely ambiguous—S&P 500 returns, the Federal Funds Rate, house prices, and nominal exchange rates—in order to answer the question of which type of uncertainty, macroeconomic or financial, dominates over the U.S. business cycle. The question was inherently domestic, and the unclassified block served as an agnostic holding area for domestic series that might plausibly belong to either of the two anchored blocks. The present project repurposes the same statistical device to answer a fundamentally different, and uniquely small-open-economy, question: through which channel do external shocks enter the domestic economy? In our application, the unclassified block is populated entirely by external drivers—the five U.S. series, four Chinese series, four global indicators (including the Baker, Bloom and Davis, 2016 Economic Policy Uncertainty indices, the Caldara and Iacoviello, 2022 Geopolitical Risk indices, and the Poilly and Tripier, 2025 Regional Trade Policy Uncertainty measure), and three U.S.–China bilateral measures—while Taiwan's own domestic variables are all pre-classified. In contrast to Davidson, Hou and Koop (2025)'s use of the device for domestic classification, $h_{m,t}$ and $h_{f,t}$ in our application are anchored by 19 clearly classified macroeconomic variables and 10 clearly classified financial variables, ensuring that the two common factors retain a well-defined economic interpretation as Taiwan's domestic macroeconomic and financial uncertainty. The 16 external drivers then load onto these anchored factors in proportions determined entirely by the data, yielding a direct reading of the operative transmission mechanism at each point in time. This reorientation transforms the classification tool of Davidson, Hou and Koop (2025) into a transmission-channel identifier for external shocks—

a substantive methodological contribution that goes beyond a country replication exercise and provides a template for applying the OI-SVMVAR framework to any small open economy exposed to competing sources of external uncertainty.

(1.3) Order-Invariance and the Incompatibility of Block Exogeneity

The identification strategy introduced in subsection (1.1)—an unrestricted contemporaneous impact matrix $\mathbf{B}_0$ whose off-diagonal elements are all freely estimated—represents a deliberate departure from the recursive Cholesky scheme that remains the workhorse of large-VAR applications. Under a Cholesky decomposition, $\mathbf{B}_0$ is constrained to be lower triangular, so that variable k may be contemporaneously affected only by variables $1, \ldots, k-1$. This recursive structure imposes an ordering on the elements of $\mathbf{y}_t$ that has no economic justification: for an N-variable system there are $N!$ possible orderings, and in a 45-variable model this amounts to more than 10^{55} admissible permutations, none of which is privileged by economic theory. Davidson, Hou and Koop (2025) document empirically that, in large-scale Cholesky-based SVMVARs, the resulting impulse response functions and forecast error variance decompositions depend substantively on whether a given variable is placed early or late in the ordering, with the same shock producing quantitatively distinct responses across permutations. This arbitrary sensitivity to ordering motivates their order-invariant alternative, whose mechanics—because zero restrictions on $\mathbf{B}_0$ interact with it in a non-obvious way—merit formal exposition before we explain why the standard small open economy (SOE) restriction of block exogeneity cannot be layered on top.

The order-invariant sampler of Davidson, Hou and Koop (2025) resolves the ordering problem through a reparameterization of the rows of $\mathbf{B}_0$. Let $\mathbf{b}_{0,i}$ denote the i-th row of $\mathbf{B}_0$, treated as a column vector of dimension N, and recall that the normalization $B_{0,ii} = 1$ removes one degree of freedom, leaving $N-1$ free parameters per row. Davidson, Hou and Koop (2025) introduce the auxiliary linear transformation

$$\mathbf{w}_i = \mathbf{d}_i + \mathbf{Q}_i \, \mathbf{b}_{0,i}, \tag{6}$$

where $\mathbf{d}_i$ is a known offset vector of dimension $N-1$ and $\mathbf{Q}_i$ is an $(N-1) \times N$ matrix constructed so that the conditional posterior of $\mathbf{w}_i$—integrating over all possible orderings of the N variables—is a standard multivariate normal, $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{N-1})$, that is invariant to the permutation applied to the variable indices. Draws from the posterior of $\mathbf{b}_{0,i}$ are then obtained by drawing $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{N-1})$ and applying the inverse transformation $\mathbf{b}_{0,i} = \mathbf{Q}_i^{\dagger}(\mathbf{w}_i - \mathbf{d}_i)$, where $\mathbf{Q}_i^{\dagger}$ is the Moore–Penrose pseudoinverse of $\mathbf{Q}_i$. Because the same conditional posterior obtains for any permutation of the variable indices, inference from the resulting MCMC sampler is order-invariant by construction: impulse responses, variance decompositions, and classification probabilities are identical regardless of the order in which the 45 variables enter $\mathbf{y}_t$.

This machinery resolves the identification obstacle introduced by a large information set, but it does so under a specific structural assumption—that $\mathbf{b}_{0,i}$ is a free vector in $\mathbb{R}^N$, subject only to the diagonal normalization. Any attempt to superimpose the zero restrictions that underlie block exogeneity disrupts this structure. In conventional SOE VARs, researchers routinely partition the variables into an external block (in our context, the U.S., Chinese, and global variables) and a domestic block (Taiwan), and then impose that the external block is permitted to affect the domestic block but not vice versa. Mechanically, this is operationalized as zero restrictions on the rows of $\mathbf{B}_0$ corresponding to external variables: for each external variable i, all elements of $\mathbf{b}_{0,i}$ corresponding to domestic variables are set to zero. If K_i such zero restrictions are imposed on row i, then $\mathbf{b}_{0,i}$ is confined to a restricted subspace of dimension $N - K_i$ rather than living freely in $\mathbb{R}^N$.

These zero restrictions are mathematically incompatible with the order-invariant reparameterization in Equation (6). The transformation $\mathbf{w}_i = \mathbf{d}_i + \mathbf{Q}_i\,\mathbf{b}_{0,i}$ requires $\mathbf{Q}_i$ to be constructed so that its $(N-1)$ rows span the orthocomplement of the normalization constraint within the full N-dimensional ambient space of $\mathbf{b}_{0,i}$. When zero restrictions collapse the effective dimension of $\mathbf{b}_{0,i}$ from N to $N - K_i$, the construction of $\mathbf{Q}_i$ degenerates: the matrix maps an $(N - K_i)$-dimensional feasible set into an $(N-1)$-dimensional image space, so that $\mathbf{Q}_i$ becomes rank-deficient on the restricted domain. The inverse transformation $\mathbf{b}_{0,i} = \mathbf{Q}_i^{\dagger}(\mathbf{w}_i - \mathbf{d}_i)$ is no longer well-defined as a bijection between the posterior of $\mathbf{w}_i$ and the posterior of $\mathbf{b}_{0,i}$: generic draws from $\mathcal{N}(\mathbf{0}, \mathbf{I}_{N-1})$ will map, under the pseudoinverse, to vectors that violate the zero restrictions, while the Jacobian of the reparameterization, $|d\mathbf{w}_i/d\mathbf{b}_{0,i}| = |\mathbf{Q}_i|$, is either zero or ill-defined on the restricted domain. The conditional posterior from which the sampler draws therefore fails to correspond to the correctly-normalized posterior of the restricted $\mathbf{b}_{0,i}$, and the order-invariance property—which depends critically on the permutation symmetry of the full N-dimensional parameter space—is broken by the introduction of direction-specific zeros. In short, the combination of block exogeneity and order invariance cannot be achieved by a superficial modification of the DHK sampler: recovering valid posterior inference under zero restrictions would require a complete redesign of the Gibbs step for $\mathbf{B}_0$, including alternative reparameterizations, restriction-aware priors, and new proposal distributions—amounting to a different algorithm altogether.

The appropriate response is not to abandon order invariance, but to replace the zero-restriction approach with the unclassified variables mechanism described in subsection (1.2). By placing all 16 external variables in the unclassified block, this project imposes no zero restrictions on any row of $\mathbf{B}_0$: the full DHK MCMC algorithm is preserved without modification, and the Q-transform in Equation (6) remains valid for every row of the contemporaneous impact matrix. Crucially, this does not mean that the SOE assumption is discarded; rather, it is relocated from a prior restriction to a post-estimation diagnostic. The economic claim that Taiwan's domestic macroeconomic and financial conditions do not systematically drive U.S. monetary policy or Chinese industrial production is assessed ex post via the forecast error variance decomposition described in subsection (2.2): if the share of forecast error variance of each external variable attributable to Taiwan's two domestic uncertainty factors $h_{m,t}$ and $h_{f,t}$ is negligible across all horizons, the SOE property holds approximately in the data. Any sizeable share would indicate that the SOE restriction is counterfactual and ought not to have been imposed in the first place. This design is both methodologically principled and substantively more informative: it delivers a testable SOE assumption—one whose validity emerges from the data—rather than an untested assumption hardwired into the identification scheme.

(2) Anticipated problems and means of resolution

(2.1) Data Assembly and Preprocessing

The empirical strategy set out in subsections (1.1)–(1.3) is demanding of the data: identifying order-invariant posterior classifications for 16 external drivers against anchored domestic factors requires a balanced panel that is simultaneously broad in cross-section, long in time, and harmonized across frequency and vintage. To meet these requirements, we assemble a monthly dataset of $N = 45$ variables spanning **2000M1 through 2025M6**, comprising approximately 306 monthly observations. The dataset is partitioned into the three blocks used by the OI-SVMVAR: 19 macroeconomic variables, 10 financial variables, and 16 unclassified variables, with the complete list and source attribution reported in Table 1 of subsection (1.2). The Taiwan macroeconomic block is drawn primarily from the Directorate-General of Budget, Accounting and Statistics (DGBAS), supplemented by customs trade data from the Ministry of Finance and by business-climate surveys from the Chung-Hua Institution for Economic Research (CIER). The Taiwan financial block combines short-rate and monetary aggregate series from the Central Bank of the Republic of China (CBC), corporate- and sovereign-bond series from TEJ, and equity-market series from the Taiwan Stock Exchange (TWSE), with TAIEX realized volatility computed from daily returns. The unclassified block draws on FRED for the five U.S. series and for the VIX; on *PolicyUncertainty.com* for the Baker, Bloom and Davis (2016) Economic Policy Uncertainty indices; on the National Bureau of Statistics of China (NBS) and the People's Bank of China (PBoC, via WIND) for the four Chinese series; on Caldara and Iacoviello (2022) for the Global, Taiwan Strait, and U.S.–China bilateral Geopolitical Risk indices; on Poilly and Tripier (2025) via author replication for the U.S. Regional Trade Policy Uncertainty index; and on the U.S. Census Bureau for U.S.–China bilateral trade volume.

The sample begins in 2000M1 for two related reasons. First, Taiwan acceded to the World Trade Organization in January 2002, and the years immediately preceding accession witnessed substantial liberalization of trade and financial flows; beginning in 2000 captures this pre-accession baseline while avoiding the deeper structural break associated with Taiwan's pre-liberalization trade regime. Second, the 1997–1998 Asian financial crisis introduced a severe structural break in cross-country financial linkages across the region, so that sample periods extending into the crisis would blend two fundamentally different integration regimes. Excluding the crisis period ensures that estimated model parameters reflect the post-crisis integration structure that characterizes the sample of primary interest. The resulting sample of approximately 306 observations provides sufficient degrees of freedom for Bayesian MCMC estimation of the 45-variable model to achieve reliable posterior convergence, consistent with the simulation evidence in Davidson, Hou and Koop (2025), who document that posterior precision for common log-volatility factors improves markedly once T exceeds roughly 300 months for systems of comparable dimensionality.

All series are harmonized to a common monthly frequency and transformed to achieve stationarity before estimation. Following Davidson, Hou and Koop (2025), the transformation protocol is applied uniformly to each variable class and proceeds in seven steps. *First*, all level series—including industrial production, employment, retail sales, exports and imports, price indices, monetary aggregates, and U.S. and Chinese production indices—are expressed as 12-month log-differences (year-on-year log-changes), which removes deterministic seasonality and induces stationarity in a single step while preserving the interpretation of each observation as a growth rate. *Second*, short-term interest rate levels—in particular the overnight interbank rate and the U.S. Federal Funds Rate—are expressed as monthly first differences, since their near-unit-root behavior would otherwise render the levels non-stationary over the 2000–2025 sample. *Third*, the term spread and credit spreads are retained in levels, because they are already stationary by construction as the difference of two rates; additional differencing would over-whiten these series and eliminate the low-frequency variation that carries the financial-cycle signal. *Fourth*, realized volatility measures for the TAIEX and for the NTD/USD exchange rate are retained in levels, as they are non-negative by construction and, in the present sample, display mean-reverting behavior consistent with stationarity. *Fifth*, seasonal adjustment via X-13-ARIMA-SEATS is applied to any series not already seasonally adjusted by the source agency; DGBAS's seasonally adjusted series are accepted as published, and FRED series apply Census X-13 at source. *Sixth*, the augmented Dickey–Fuller (ADF) and KPSS tests are applied to every transformed series at the 5% significance level, with rejection under the ADF null of a unit root combined with non-rejection under the KPSS null of stationarity taken as joint evidence of stationarity. Any series that fails to attain stationarity under both tests receives an additional first difference and is retested; this safeguard ensures that the balanced panel entering the OI-SVMVAR is stationary in the pre-estimation sense required by the Bayesian algorithm. *Seventh*, all transformed series are standardized to zero mean and unit variance prior to estimation, following Davidson, Hou and Koop (2025), so that the prior distributions on the VAR coefficient matrices $\mathbf{A}_l$ and on the factor loadings in $\mathbf{\Gamma}$ are scale-invariant across variables with widely different units and levels of variance.

Missing observations arise occasionally for a small number of Chinese series in the early months of the sample, where monthly publication practice lagged international conventions until the mid-2000s. For each affected series, we apply linear interpolation when the gap is at most two consecutive observations; for longer gaps, the sample start date for the full panel is trimmed to the earliest month at which all 45 series are observed. The complete record of each raw series, its source identifier, the transformation applied, the stationarity test results, and any interpolation decisions is maintained in a replication log and accompanying codebook, ensuring that the entire data-construction pipeline is reproducible from the original public-source downloads through the balanced panel that enters Equation (1).

(2.2) Works Planned for the First Year

The first year of the project is organized around four overlapping activities that move the empirical framework from raw data to preliminary posterior results. *The first activity*, spanning approximately months 1–4, is the assembly and verification of the dataset. All 45 variables are downloaded from the sources enumerated in Section (2.1), harmonized to a common monthly frequency, and assembled into the balanced panel spanning 2000M1–2025M6. Missing observations—concentrated in a small number of Chinese series during the early sample—are addressed via linear interpolation or, when necessary, by trimming the sample start date. The transformation protocol described in Section (2.1) is applied to each series, and the ADF and KPSS stationarity tests are executed and logged. The complete dataset, the transformation decisions, and the source documentation are recorded in a replication log that accompanies the final panel.

The second activity, spanning approximately months 3–7 and concurrent with the final phase of data assembly, is the implementation and adaptation of the MCMC algorithm. The order-invariant SVMVAR sampler of Davidson, Hou and Koop (2025) is implemented and adapted to the 45-variable Taiwan dataset. The algorithm proceeds as a Gibbs sampler, cycling through the following blocks: the VAR coefficient matrices $\mathbf{A}_l$ in Equation (1); the contemporaneous impact matrix $\mathbf{B}_0$ in Equation (2), sampled via the $\mathbf{Q}$-matrix reparameterization in Equation (6); the uncertainty factor loadings $\alpha_{i,m}$ and $\alpha_{i,f}$ embedded in $\mathbf{\Gamma}$; the common log-volatility paths $h_{m,t}$ and $h_{f,t}$ in Equation (4), drawn in a single block via the precision sampler of Chan and Jeliazkov (2009) as extended by Cross et al. (2023); and the Markov-switching classification probabilities $\pi_{i,t}$ in Equation (5), obtained via the Hamilton (1989) filter applied to the $s_{i,t}$ indicators. Convergence is assessed through the spectral diagnostic of Geweke (1992), visual inspection of MCMC trace plots, and effective sample size calculations. Estimation is conducted on the university high-performance computing cluster.

The third activity, spanning approximately months 5–9, is preliminary estimation and benchmarking. Upon completion of the sampler, a preliminary run is conducted on a reduced 30-variable model—retaining all 19 Taiwan domestic variables together with a subset of the most important external drivers—to validate the MCMC implementation and to benchmark the estimated $h_{m,t}$ and $h_{f,t}$ paths against the published results of Davidson, Hou and Koop (2025) for the U.S. economy. Upon successful validation, the full 45-variable model is estimated. Each MCMC run requires approximately 30 hours at 50,000 post-burn-in draws, based on the computational requirements reported by Davidson, Hou and Koop (2025) for a model of comparable scale.

The fourth activity, spanning approximately months 9–12, initiates the three-step analysis proper. Posterior draws from the full model are used to extract time-series of the classification probabilities $\pi_{i,t}$ for all 16 external variables, with preliminary plots produced for the key episodes identified in Section 1 (the 2008–2009 Global Financial Crisis, the 2015 China slowdown, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 shock, and the 2022 Federal Reserve tightening cycle). A preliminary forecast error variance decomposition is computed to assess the relative contributions of U.S., Chinese, and global uncertainty shocks to Taiwan's domestic factors $h_{m,t}$ and $h_{f,t}$, and preliminary findings are presented at a domestic or international conference in the final quarter of Year 1. Throughout the year, the project research assistant receives structured training in Bayesian econometrics and MCMC methods, building the technical capacity required for the full three-step analysis and for the conditional structural modelling work of Year 2.

1.2 Anticipated results and achievements

The Second Year

1.3 Methods, procedures, and implementation schedule

(1) Research principles, methods, and the innovation of research methods

(1.1) Steps 1–3: Full Three-Step Empirical Analysis (OI-SVMVAR)

The second year completes the three-step empirical analysis initiated in Year 1. Having obtained converged posterior draws from the 43-variable OI-SVMVAR, we conduct the following analyses, all computed over the joint posterior distribution so that parameter and classification uncertainty are fully propagated.

Step 1: Time-varying classification. For each unclassified external variable i, we extract the posterior time path of the classification probability $\pi_{i,t} = P(s_{i,t} = \text{macro} \mid \mathcal{F}_t)$ defined in Equation (5). Plotting $\pi_{i,t}$ across the sample period reveals *when* a given external shock source—such as U.S. monetary policy uncertainty, China's industrial production volatility, or the VIX—transmits to Taiwan through the macroeconomic channel ($\pi_{i,t} \approx 1$) versus the financial channel ($\pi_{i,t} \approx 0$). We pay particular attention to regime shifts around five landmark episodes: the 2008–2009 global financial crisis, the 2012 China–Japan territorial dispute, the 2018–2019 U.S.–China trade war, the 2020 COVID-19 pandemic, and the 2022 Pelosi visit to Taiwan.

Step 2: Forecast error variance decomposition (FEVD). We decompose the forecast error variance of Taiwan's macroeconomic uncertainty ($h_{m,t}$) and financial uncertainty ($h_{f,t}$) into contributions from each variable in the system. This quantifies the *relative importance* of each external shock source. The FEVD also serves as an ex post validation of the small open economy assumption: if Taiwan's domestic variables explain a negligible share of the variance of U.S. and Chinese indicators, the assumption that external variables are approximately exogenous is supported without having imposed it through zero restrictions on $\mathbf{B}_0$.

Step 3: Impulse response functions (IRF). We compute generalized impulse response functions of Taiwan's domestic variables to one-standard-deviation innovations in $h_{m,t}$ and $h_{f,t}$, as well as to identified shocks in the key external variables. This step reveals the *economic consequences* of uncertainty transmitted through each channel. The shape, magnitude, and persistence of these IRFs constitute the empirical targets that discipline the structural modeling in the second component of Year 2.

(1.2) Conditional Structural Modeling: A Competing Hypotheses Framework

A central challenge in writing a multi-year proposal with an "empirics-first, theory-second" design is that the structural model in Year 2 must be specified before the Year 1 results are known. We resolve this tension by adopting a **competing hypotheses framework** with **conditional extension rules**: rather than committing ex ante to a single, heavily parameterized model, we pre-specify a parsimonious baseline and a set of theoretically motivated candidate mechanisms, each linked to a distinct empirical signature that the Year 1 OI-SVMVAR can either confirm or reject. This approach, standard in top-tier empirical macroeconomics (Fernández-Villaverde and Guerrón-Quintana, 2020), ensures that the Year 2 model is disciplined by the data rather than designed to accommodate any conceivable pattern.

Baseline model. The starting point is a standard small open economy (SOE) New Keynesian model with Rotemberg (1982) quadratic price adjustment costs, habit persistence in consumption, and a Taylor-type monetary policy rule. We adopt Rotemberg pricing rather than the more common Calvo (1983) specification because Oh (2020) demonstrates that the two frameworks—while equivalent for first-moment shocks—generate qualitatively opposite predictions for uncertainty shocks: the Calvo model produces counterfactual inflation increases through precautionary pricing, whereas the Rotemberg model correctly predicts the deflationary response observed in VAR evidence. The baseline model is augmented with exogenous stochastic volatility in the productivity process, following Fernández-Villaverde et al. (2011), and solved via third-order perturbation in Dynare, which is the minimum approximation order at which uncertainty shocks enter the policy functions (Fernández-Villaverde and Guerrón-Quintana, 2020).

Competing hypotheses. The international literature on uncertainty transmission has identified four primary channels through which external uncertainty shocks propagate to the real economy. Each channel implies a distinct structural mechanism and, crucially, a distinct empirical signature in the Year 1 IRFs. We now describe each hypothesis, its theoretical foundation, the Year 1 trigger that would activate it, and the corresponding model extension.

Hypothesis 1: The real "wait-and-see" channel. Bloom (2009) shows that when future uncertainty is high, firms facing irreversible investment decisions optimally postpone capital expenditure and hiring until the uncertainty dissipates. This channel operates through the real side of the economy and does not require any financial market imperfection. Carrière-Swallow and Céspedes (2013) extend this argument to 40 countries and find that developed economies with deep financial markets exhibit the classic Bloom (2009) pattern: a sharp but transient investment drop followed by a rapid overshoot recovery.

Year 1 trigger: The OI-SVMVAR IRFs show that Taiwan's private investment and durable goods consumption decline sharply following an external uncertainty shock, but corporate credit spreads and bank lending rates do not respond significantly. The classification probabilities $\pi_{i,t}$ for external variables remain persistently near unity (macroeconomic classification).

Conditional model extension: We introduce non-convex capital adjustment costs that generate an inaction region (à la Bloom (2009)), combined with search-and-matching frictions in the labor market. Financial frictions are **excluded** from the model. The testable prediction is that shutting down the investment adjustment cost eliminates the contractionary effect of uncertainty shocks.

Hypothesis 2: The financial accelerator channel. Christiano, Motto and Rostagno (2014) and Gilchrist, Sim and Zakrajšek (2014) argue that uncertainty shocks widen the information asymmetry between borrowers and lenders, causing risk premia to spike and credit supply to contract. In the Bernanke–Gertler–Gilchrist (BGG) framework, the external finance premium rises endogenously with uncertainty, amplifying the initial shock through a financial accelerator mechanism. For small open economies, Carrière-Swallow and Céspedes (2013) provide cross-country evidence that financial market depth is a key determinant of the transmission amplitude, and estimate that the credit channel accounts for up to one-half of the excess investment decline in emerging economies.

Year 1 trigger: Credit spreads widen significantly and equity prices decline *before* real activity contracts. The time-varying classification probabilities shift external variables toward the financial block ($\pi_{i,t} \approx 0$). The FEVD assigns a substantial share of $h_{f,t}$ variance to external shock sources.

Conditional model extension: We augment the baseline with a BGG-type financial accelerator in which entrepreneurs face a costly state verification problem. The key parameter is the elasticity of the external finance premium with respect to leverage. Pure real rigidities (non-convex adjustment costs) are **excluded**. The testable prediction is that the model replicates the leading response of credit spreads observed in the VAR.

Hypothesis 3: The balance sheet and aggregate risk concentration channel. Di Tella (2017) provides a distinct financial mechanism: uncertainty shocks—specifically, increases in idiosyncratic risk—endogenously concentrate aggregate risk on the balance sheets of leveraged financial intermediaries. Unlike the BGG framework, this channel survives even under complete financial contracts, because uncertainty alters intermediaries' optimal risk-taking behavior through an income effect. The quantitative implications are substantial: when intermediaries' balance sheets deteriorate, asset prices fall, risk premia rise, and investment contracts through an amplification loop that is absent under standard TFP shocks.

Year 1 trigger: Financial sector variables (bank equity, financial institution CDS spreads) react more strongly than non-financial corporate credit spreads. Foreign portfolio outflows are a leading indicator of the domestic downturn. The classification probabilities for VIX and U.S. credit spreads shift decisively toward the financial block.

Conditional model extension: We introduce a banking sector with leveraged intermediaries who must retain equity (skin-in-the-game constraint). The key departure from Hypothesis 2 is that the friction operates on the *intermediary's* balance sheet rather than the *borrower's*. The testable prediction is that the model generates a strong co-movement between intermediary leverage and asset price declines that is absent in the BGG specification.

Hypothesis 4: The exchange rate and imported inflation channel. This hypothesis is specific to small open economies. When external uncertainty spikes, global capital flows toward safe-haven currencies (primarily the U.S. dollar), causing the SOE currency to depreciate sharply. The depreciation raises import prices, generating inflationary pressure precisely when the domestic economy is contracting—a "stagflation" pattern that forces the central bank into a policy dilemma. Mumtaz and Theodoridis (2015) provide empirical support using a nonlinear SOE DSGE model estimated on U.S.–U.K. data, finding that U.S. supply-type volatility shocks transmit internationally and generate exactly this negative co-movement between output and inflation in the recipient economy.

Year 1 trigger: The New Taiwan dollar depreciates significantly, and domestic CPI rises (or fails to decline) despite falling output—a pattern inconsistent with a pure demand shock. The classification probabilities for the Federal Funds Rate and U.S. EPU oscillate between macro and financial blocks, reflecting the dual nature of the exchange rate channel.

Conditional model extension: We extend the baseline with incomplete exchange rate pass-through to import prices, a UIP risk premium shock, and producer currency pricing for Taiwan's exports. The monetary policy rule is augmented with an exchange rate stabilization motive, following the empirical evidence on CBC intervention behavior. The testable prediction is that the model generates the stagflationary pattern—simultaneous output contraction and CPI increase—that would be absent in a closed-economy specification.

(1.3) From Identification to Discrimination: The Year 1–Year 2 Bridge

The value of the competing hypotheses framework lies not in its ability to match any given set of IRFs—a sufficiently parameterized model can always achieve high fit—but in its ability to **discriminate** between mechanisms by exploiting qualitative differences in their predictions. We now formalize the decision protocol that links the Year 1 empirical output to the Year 2 model selection.

The four hypotheses generate distinct, falsifiable predictions along three diagnostic dimensions that the Year 1 OI-SVMVAR directly estimates: (i) the *relative timing* of financial versus real variable responses in the IRFs; (ii) the *sign* of the inflation response (deflationary under Hypotheses 1–3 versus stagflationary under Hypothesis 4); and (iii) the *time-varying classification* pattern of external variables (persistently macroeconomic under Hypothesis 1, persistently financial under Hypotheses 2–3, and time-varying under Hypothesis 4).

We emphasize that empirical reality may not correspond neatly to a single hypothesis. The Year 1 results may reveal that different external shock sources operate through different channels, or that the dominant channel shifts across historical episodes. In such cases, we will construct a model that nests the two most empirically relevant mechanisms and test whether the data can discriminate between them using posterior model comparison (marginal likelihood). This "horse race" approach, advocated by Fernández-Villaverde and Guerrón-Quintana (2020), is strictly preferable to building a kitchen-sink model that includes all four mechanisms simultaneously, because it maintains the parsimony needed for sharp identification.

To ground this discriminatory exercise in the existing literature, we note that Born and Pfeifer (2021) provide a cautionary benchmark: they show that the New Keynesian markup channel—often invoked as the primary propagation mechanism for uncertainty shocks—is largely inconsistent with the data for price markups and generates quantitatively small output effects. This finding reinforces our commitment to letting the Year 1 evidence, rather than theoretical priors, determine the model structure. We will explicitly test whether the markup channel contributes to uncertainty propagation in the selected specification, and report the results regardless of their sign.

The structural model, once selected and estimated, will be used for two purposes. First, **counterfactual simulations**: we shut down individual transmission channels (e.g., setting the financial accelerator elasticity to zero) and quantify how much of the empirical IRF each channel explains. Second, **optimal policy analysis**: conditional on the identified dominant channel, we evaluate whether the Central Bank of China (Taiwan) should prioritize interest rate stabilization, exchange rate intervention, or macroprudential measures when facing external uncertainty shocks.

(2) Anticipated problems and means of resolution

(2.1) Computational and Estimation Challenges

The Year 2 research program faces three computational challenges. First, the full 43-variable OI-SVMVAR requires approximately 30 hours per MCMC run on a modern multi-core workstation; we will secure access to the university's high-performance computing cluster and run multiple chains in parallel to ensure convergence. Second, the nonlinear SOE DSGE model must be solved via third-order perturbation, which is computationally feasible in current versions of Dynare (version 6+) but requires careful pruning of explosive higher-order terms (Fernández-Villaverde and Guerrón-Quintana, 2020). Third, Bayesian estimation of the DSGE model conditional on the

Year 1 uncertainty factors requires a two-step procedure: the posterior means of $h_{m,t}$ and $h_{f,t}$ from the OI-SVMVAR serve as observable inputs to the DSGE likelihood, following the approach of Mumtaz and Theodoridis (2015) who use a similar empirical-then-structural estimation strategy.

(2.2) Works planned for the second year

The second-year work plan proceeds in four phases. In the first quarter (months 13–15), we complete the full 43-variable OI-SVMVAR estimation and conduct the three-step empirical analysis (classification probabilities, FEVD, and IRFs), including robustness checks comparing the small-scale (30-variable) and large-scale (43-variable) specifications. In the second quarter (months 16–18), we evaluate the Year 1 IRFs against the four diagnostic criteria outlined in subsection (1.3), select the one or two most empirically supported hypotheses, and construct the corresponding SOE DSGE model. In the third quarter (months 19–21), we estimate the structural model via Bayesian methods, conduct counterfactual simulations, and perform the optimal policy analysis. In the final quarter (months 22–24), we complete the research paper, prepare a policy brief for the Central Bank of China (Taiwan), and submit to a top-tier international journal.

1.4 Anticipated results and achievements

2 Integrated research project

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